

Simple Belief Elicitation

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Abstract We present a method for eliciting beliefs about probabilities and quantiles when multiple realisations of an outcome are available. The methods are applicable for any reasonable utility function. Unlike existing techniques that account for deviations from risk-neutrality, these methods are highly transparent to subjects and easy to implement. Rather than identifying point beliefs these methods identify bounds on beliefs, thus trading off precision for generality and simplicity.

Keywords Belief elicitation, experimental methodology, median elicitation, scoring rule

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1 Introduction

Experimental economists are increasingly recognising the value of directly eliciting the beliefs of their subjects. For example, direct measures of a subject's beliefs can help us disentangle whether deviations from homo economicus behaviour is due to social preferences or bounded rationality, test whether belief updating is Bayesian, and learn about whether peer effects are caused by imitation or information transmission.

There are, by now, a large variety of incentive compatible methods for eliciting beliefs, with various strengths and weaknesses.¹ Some of the most commonly used are not incentive compatible for risk-averse subjects, (e.g. linear or quadratic scoring rules) while those that do not suffer from this flaw tend to be either time-intensive, (e.g. calibrating elicited beliefs; Offerman et al, 2009) or challenging for subjects to fully understand (e.g., variations of the Becker-DeGroot-Marschak mechanism; Karni, 2009). For the working experimentalist, the task of selecting and implementing a method can be intimidating. Here we present procedures for eliciting beliefs about probabilities as well as means, variances and quantiles of distributions, that are robust to risk-aversion, require minimal labtime, and are simple for subjects to understand. While these methods require multiple realizations of the probabilistic event, and return bounds on the desired statistic rather than a point estimate, they are straight-forward to implement and will be suitable for the purposes of many researchers.

Most of the literature on belief elicitation focuses on payments based on the actual outcome of a single event. However, in many laboratory experiments, there will be not just one but many independent realisations of the random variable of interest. Take, for example, a one-shot prisoners' dilemma experiment where the experimenter is interested in beliefs the subjects hold about the probability of defection. If there are 20 subjects per session, each stated belief can be matched with the 19 realizations of the decisions of others. The two methods we discuss in this paper, the first for eliciting probabilities, the second for quantiles, take advantage of these multiple realisations. In doing so we remove the need to refer in experimental instructions to numerical probabilities of single events, which many subjects may have difficulty understanding (see section 5 for evidence of this).

In order to elicit information regarding the probability of different outcomes, the subject is asked to guess the empirical frequency of each outcome. A prize is then awarded if and only if their guess coincides with the realized frequencies. For the case of only two outcomes, this method has been used before (Wilcox and Feltovich, 2000; Bhatt and Camerer, 2005; Hurley and Shogren, 2005; Costa-Gomes and Weizsacker, 2008; Blanco et al, 2010; Le Coq et al, 2015), however its properties do not appear to have been well understood by the experimental community. Wilcox and Feltovich (2000) and

¹ For detailed and comprehensive discussions see Schlag et al (2015) and Schotter and Trevino (2014).

Blanco et al (2010) state only that beliefs about the modal frequency of outcomes are elicited, while Costa-Gomes and Weizsacker (2008) say that it is valid only when the true subjective probability coincides exactly with one of the possible empirical distributions. For this special case of binary outcomes, a correct interpretation was reported in Hurley and Shogren (2005) but not given much prominence in the paper, and as a result appears to have been largely overlooked.²

Not only does this method elicit beliefs about modal frequencies, but we show that it also enables the researcher to identify a region on a simplex in which the belief of the subject should lie. Inference does not require postulating any assumptions on the utility function beyond assuming that the subject strictly prefers the prize. For binary events this region is an interval of width $1/(n+1)$, where n is the number of realizations of the variable in question. For example, for $n = 19$, such as in the prisoners' dilemma example given above, the size of the interval is 5%. Given that subjects tend to answer questions about percentages in multiples of five (Manski, 2004), this would involve no practical loss of precision. We prove the result for events with an arbitrary number of possible outcomes and draw attention to features that make it attractive for practical belief elicitation, such as avoiding complex experimental instructions. We also show that, for binary outcomes, it is most precise in a well defined sense. Responses elicited using this method can be used to estimate bounds on subjects' beliefs about means and variances of distributions.

We also propose a related method using multiple realisations of a random variable to elicit beliefs about the median of a distribution. Subjects are asked to report a number such that half the realizations will be below that number, and are rewarded the prize if and only if this condition holds. As with the first method, this is extremely straightforward to explain to subjects, and is equally valid for all non-trivial utility functions. Similar techniques work for other quantiles, and thus for measures of spread such as the interquartile range.

The paper proceeds as follows: sections 2 and 3 describe the theory underlying the methods for eliciting probabilities and quantiles, respectively; section 4 gives practical examples of how each method could be implemented in typical experiments, first in a Trust game, then in a Public Goods game; in section 5 we discuss the strengths and weaknesses of the methods in detail, while section 6 concludes.

2 Eliciting Probabilities and Functions Thereof

In this section we present a method for eliciting probabilities, derive tight bounds on the “true” underlying probabilities and show how this can be used to gain an understanding of perceived means, variation and expected utility.

² Another reason the method under discussion may have been largely disregarded is that the main message of the paper is that they failed to recover induced beliefs. However, on closer inspection, this failure is attributed to the induction process rather than the elicitation method.

2.1 Eliciting Probabilities

Let Y be a random variable with k possible outcomes $s_1, \dots, s_k$, where p_i is a subject's subjective belief about the probability that outcome s_i will occur. Subjects are asked to report $b = (b_1, \dots, b_k)$, b_i being a non-negative number for all i , and are paid a prize of value R if and only if for all $i \in \{1, \dots, k\}$ b_i is equal to the number of times s_i occurs out of n independent³ realisations of Y . We call this the frequency guessing method. In the context of a strategic form game in a laboratory experiment, each s_i is a strategy available to the subject's partner. The subject is then told that they will be awarded a prize if they can correctly guess the number of people in their partner's role who play each strategy. Thus, from the standpoint of the subject making the report, the prize will be awarded with probability

$$f(b) = \frac{n!}{b_1! \cdot \dots \cdot b_k!} \prod p_i^{b_i}.$$

It follows immediately that the subject maximizes expected utility if and only if they maximize the probability f of receiving the prize. Hence, and without loss of generality, we are interested in the relationship between the maximizers of f and the underlying subjective beliefs. In the following we provide a complete characterization of this relationship.

Let B be the set of feasible reports, so

$$B = \left\{ b \in \{0, 1, \dots, n\}^n : b_i \geq 0 \forall i, \sum_{i=1}^n b_i = n \right\}.$$

Proposition 1 *Consider $b \in B$. Then b maximises f over all B if and only if*

$$\begin{aligned} \frac{b_i}{b_j + 1} &\leq \frac{p_i}{p_j} \leq \frac{b_i + 1}{b_j} \quad \forall j \neq i \text{ when } p_j, b_j \neq 0 \\ b_j &= 0 \text{ if } p_j = 0. \end{aligned} \quad (1)$$

In particular, if b maximizes f then

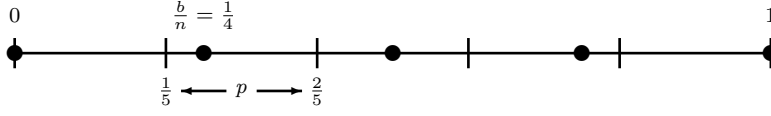
$$\frac{b_i}{n + k - 1} \leq p_i \leq \frac{b_i + 1}{n + 1} \text{ holds for all } i. \quad (2)$$

Proof See online appendix.

Figure 1 demonstrates this result for $k = 2$ and $n = 4$. The dots show the possible reports (divided by n) and the surrounding intervals show the possible values of p given the reports. In the figure we see that only those beliefs on the boundary between two regions give rise to two different optimal reports. More generally, our proof of Proposition 1 reveals that any subject with beliefs that satisfy (1) with strict inequalities has a unique best report.

³ The independence assumption will be discussed in Section 5.

⁴ For the special case of $k=2$, this result appears in Hurley and Shogren (2005).

Fig. 1 Reported and consistent true beliefs for $k = 2$ and $n = 4$ 

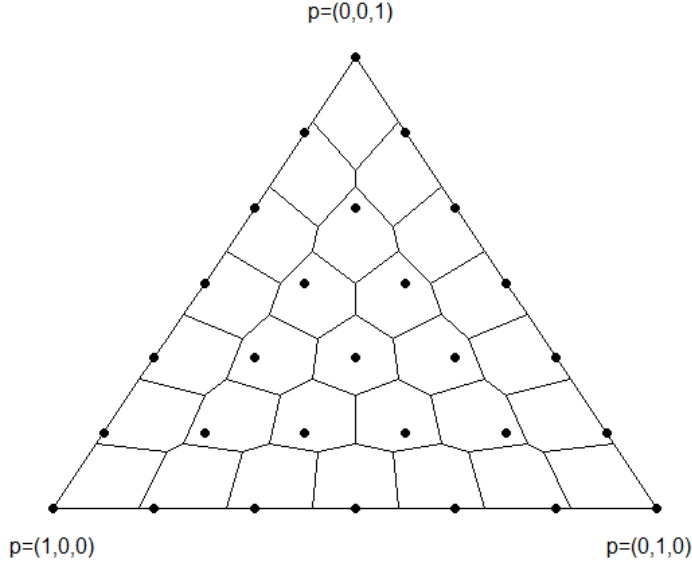
For $k = 2$ we hasten to point out that one cannot extract more precise information for any given utility function in the following sense. Consider any alternative payment rule with the same input, that is a subject's stated belief about the number of times that an outcome will occur. For a given utility function u let P_b^u be the set of beliefs under which it is optimal under the alternative rule to report b , $b \in \{0, 1, \dots, n\}$. Then $\cup_{b \in \{0, 1, \dots, n\}} P_b^u = [0, 1]$. Consequently, $\max_{b \in \{0, 1, \dots, n\}} d(P_b^u) \geq 1/(n+1)$ where $d(P_b^u)$ is maximal distance between any two points belonging to P_b^u (where d is its width if P_b^u is an interval). Let the *minimal precision* of a rule be the negative of the maximal difference between any two probabilities that lead to the same report. Then we find that there is no payment rule with a strictly higher minimal precision than the one we have presented. In fact, it is easy to see that the inferred true probabilities of any rule with this value of minimal precision are unique. We summarize.

Proposition 2 *Any alternative rule that elicits the frequency of the occurrence of a single event (so $k = 2$) has a strictly lower minimal precision than that of the frequency guessing method.*

In general the set of feasible probabilities is constrained by $p_i \geq 0$ for all i , by $\sum_{i=1}^n p_i = 1$ and by the constraints given in (1). Figure 2 shows how these constraints divide the simplex into regions of feasible combinations of "true" beliefs given each report, for $k = 3$ and $n = 6$.

2.2 Eliciting mean and variance

The bounds on probabilities derived in the previous subsection can be used to also place bounds on beliefs about the mean and variance of a distribution.

Fig. 2 Reported and consistent true beliefs for $k = 3$ and $n = 6$ 

Assume that $s_i \in \mathbb{R}$ for $i = 1, \dots, k$. If one has elicited $\{b_i\}_{i=1}^k$ then one can derive 100% confidence intervals for functions of Y , such as EY and $VarY$.

Consider for instance EY . We obtain $EY \in [L(b), U(b)]$ where

$$L(b) = \min \left\{ \sum_{i=1}^k p_i s_i \text{ s.t. } p_i \geq 0, \sum p_i = 1, \text{ and } b_j p_i \leq (b_i + 1) p_j \forall i \neq j \right\}$$

and

$$U(b) = \max \left\{ \sum_{i=1}^k p_i s_i \text{ s.t. } p_i \geq 0, \sum p_i = 1, \text{ and } b_j p_i \leq (b_i + 1) p_j \forall i \neq j \right\}.$$

The width of this interval, $U(b) - L(b)$, depends on the precise values of the stated beliefs. Bounds are tight given that the underlying constraints as stated in Proposition 1 are necessary and sufficient.⁵ Table 1 shows bounds on beliefs about means for $k = 3$, $n = 10$, and a selection of stated beliefs when $s_i = \frac{i}{3}$ for $i \in \{1, 2, 3\}$. Bounds on beliefs about variances can be calculated in a similar way and are shown in the same table.

Precision is decreasing in k and increasing in n . For example for $k = 5$ and $n = 10$ the maximal interval width when eliciting information about a mean

⁵ The same calculations also give the maximum and minimum possible expected utility of a subject playing a particular strategy when s_i is the payoff to the subject associated with that strategy when their partner plays strategy i .

Table 1 Maximum and minimum means and variances for $k = 3$ and $n = 10$

Stated belief	Max Mean	Min Mean	Δ Mean	Max Var	Min Var	Δ Var
(3,3,4)	0.727	0.667	0.061	0.083	0.070	0.013
(2,3,5)	0.788	0.722	0.066	0.077	0.062	0.015
(1,1,8)	0.909	0.833	0.076	0.068	0.042	0.026
(0,0,10)	1	0.917	0.083	0.039	0	0.039

is 0.138, whereas for $k = 3$ and $n = 20$ it is 0.045. For $k = 5$ and $n = 20$ the maximal interval width is 0.083.

3 Eliciting the Median and Quantiles

3.1 Median

In the following we present a method, the *median guessing method*, for eliciting the median of the belief distribution of a subject. The method determines when to award a prize based on the report of the subject and on n realizations of the underlying random variable Y .

We consider only the case where n is even. While all results extend in a straightforward way to odd n , in this case the instructions for subjects become less intuitive. For even n the instructions would be along the following lines: “Please tell us a number such that you think half the outcomes will be below or equal to that number. If your guess is correct you will receive R .” Odd n requires explaining that the prize will be paid if either $\frac{n-1}{2}$ or $\frac{n+1}{2}$ outcomes are weakly less than the guess. There is no advantage to this increased complexity⁶ so when the maximal number of subjects in the role of the subject’s partner is odd we suggest to drop one of the realizations at random.

Recall that the a real number $m_{1/2}$ is called a *median* of the random variable Y if $P(Y \leq m_{1/2}) = P(Y \geq m_{1/2}) = 1/2$. There are several different ways do adjust the definition when there is no real number $m_{1/2}$ with this property. In the following we define the median of Y equal to $m_{1/2}$ if $m_{1/2}$ is the smallest number m that satisfies $P(Y \leq m) \geq 1/2$. With this definition, the median both always exists and is unique due to the right continuity of any cdf.

According to the median guessing method, the subject is asked to report $m \in \mathbb{R}$ and is awarded a prize with value R if and only if half the realizations are smaller than or equal to m . Thus the prize is awarded with probability

$$f(m) = \binom{n}{n/2} P(Y \leq m)^{n/2} P(Y > m)^{n/2}.$$

Analogous to Section 2 without loss of generality we now investigate which reports maximize the probability of winning the prize. Again the only assumption on risk preferences is that $u(R) > u(0)$.

⁶ Unlike the methods in the previous section increasing n here does not increase precision.

Consider first the case where the underlying distribution has no point masses.

Proposition 3 *Assume that n is even and $P(Y = y) = 0$ for all y . Then m maximizes f if and only if m is the median of Y .*

Proof See online appendix.

Consider now the case where point masses are possible. In particular, assume that Y only takes values on a finite grid $X = \{x_1, \dots, x_K\} \subset \mathbb{R}$ with $x_i < x_{i+1}$ for all $i = 1, \dots, K - 1$. In this case it is enough to ask for reports that belong to X . Assume that each outcome occurs with strictly positive probability. We find that the report that maximizes the probability of receiving the prize is either equal to the median of their beliefs or equal to the next lower outcome.

Proposition 4 *Assume n even, $Y \in X$ with X finite and $P(Y = x) > 0$ for all $x \in X$. If $x_m \in X$ maximizes f then the median of Y is contained in $\{x_{m-1}, x_m\}$.*

Proof See online appendix.

It is easy to see that the proof also goes through if one allows for multiple medians by defining $m_{1/2}$ to be a median of Y if $P(Y \geq m_{1/2}) \geq 1/2$ and $P(Y \leq m_{1/2}) \geq 1/2$. One only has to adjust the statement of the proposition, replacing “the median” by “a median”.

We suggest to use the report x_m as an estimate of the median, and to use the pair $\{x_{m-1}, x_m\}$ for any statistical analysis.

3.2 Quantiles

Our results for median elicitation are easily extended to general quantiles. Given $q \in (0, 1)$ assume that we are interested in the quantile m_q of Y , where m_q is the smallest number that satisfies $P(Y \leq m_q) \geq q$. We show how to do this when nq is an integer. The idea is to award the prize if and only if the fraction q of the observations are smaller than or equal to m . The proofs follow analogously to those for the median, as $x^{q_1} (1 - x)^{q_2}$ is maximized over all x for given $q_1, q_2 \in \mathbb{N}$ if and only if $x = \frac{q_1}{q_1 + q_2}$.

4 Examples

In this section we give sample instructions to illustrate some of the points we have made.

4.1 Probabilities: A Trust Game

Suppose one wants to clearly identify deviations from *homo economicus* behaviour in a trust game by eliciting sender beliefs about the probability the receiver will return. An experimental session involves 11 pairs of senders and receivers: the sender can guarantee a payoff of €10 by not trusting; by trusting, the sender can receive a payoff of €15 (€0) if the receiver returns (defaults). The strategy method is used, so choices are elicited from all receivers.

Sample Instructions

- There are 10 receivers in this experiment who have made decisions with whom you have not interacted. How many of those subjects do you think chose *return*? If your guess is correct you will receive €5.

Note that the choice of the participant with whom the sender interacted is excluded to avoid hedging. Consider a (non-risk loving and selfish) sender who chose to trust, and guesses that six of the ten receivers chose to return. The true belief of this subject about the return probability of sender who guesses that six receivers chose to return is between $\frac{6}{10+2-1} = \frac{6}{11}$ and $\frac{6+1}{10+2-1} = \frac{7}{11}$. Therefore, as the expected return from trusting is less than not trusting for all probabilities in that interval, this subject cannot be best-responding to beliefs.

The bounds of the expected return for a sender who guesses that seven receivers chose to return are €9.54 and €10.91, so one cannot be sure whether trusting is consistent with expected payoff maximization. Any sender who guesses higher than seven would maximize the expected payoff by trusting.

4.2 Median and Interquartile Range: A Public Goods Game

Suppose a treatment variation leads to an increase in both the central tendency and/or spread of contributions in a standard public goods game, and one is interested in whether or not this is anticipated by subjects. The following instructions illustrate how the experiment can elicit beliefs about the median, and upper and lower quartiles of contributions.

Sample Instructions

- The contributions of two participants will be selected at random. Choose a number such that you think one of the contributions will be greater than that number, and one will be less than or equal to that number. If you are correct, you will earn an extra €5.
- The contributions of four participants (different from the two from the previous task) will be selected at random. Choose a number such that you think one of the contributions will be greater than that number, and three will be less than or equal to that number. If you are correct, you will earn an extra €5.

- The contributions of four participants (different from the six from the previous two tasks) will be selected at random. Choose a number such that you think three of the contributions will be greater than that number, and one will be less than or equal to that number. If you are correct, you will earn an extra €5.

The contributions with which the subject is matched should be different for each task to avoid hedging, so eliciting the median and two quartiles requires only ten subjects. A typical public goods game session will have at least four groups of four subjects, which also allows the experimenter to also exclude responses from the belief elicitation being matched with subjects with whom the game was played.

As the method identifies either the true belief or the true belief or the next lower value, it is best to allow the choice set of contributions in the game to be as fine as possible. For example, suppose the responses to the answers were 4, 8, and 2. If contributions were whole numbers of cents, the beliefs would be either €3.99 or €4 and the interquartile range between €5.99 and €6.01. However, if contributions were limited to multiples of €1, the equivalent numbers would be €3 or €4 for the median, and €5 and €7 for the interquartile range.

5 Discussion

Clear advantages of the methods described in this paper are that they are incentive compatible for all non-trivial utility functions, and that the ease with which they can be communicated to subjects means that one can obtain measures of beliefs without adding a great deal of time or complexity to an experiment. In the presence of heterogeneous risk-preferences, any scoring rule will result in biased responses from some subjects (see Schlag et al, 2015). Alternative methods which work for general utility functions typically require lengthy instructions, complicated payoff mechanisms, and in some cases additional tasks to approximate utility functions.

Reducing the complexity of instructions, and simplifying the communication of probabilistic information has not been a focus of the experimental economics literature on belief elicitation. Confusion and difficulties with processing probabilities without doubt increase noise and possibly introduce biases in responses. In light of this we suggest that an important route to improving the quality of belief elicitation is by better facilitating the understanding and communication of probabilities by subjects.

Probabilities can be expressed in a number of different ways: as a number, a percentage, or as a frequency. There is substantial evidence that even highly educated individuals often perceive mathematically equivalent probabilities as different when presented in the alternative formats. Lipkus et al (2001) found that in a sample where 90% of respondents had at least some tertiary education, 40% were unable to convert a percentage to a frequency, while 79% were unable to convert a frequency to a percentage. Similar but more extreme results have been found for less educated respondents (Schwarz et al, 1997).

Consequently, the format of probabilities has the potential to affect responses when eliciting beliefs.

There is evidence that people tend to be more comfortable and better able to process probabilities expressed as natural frequencies rather than other formats. Experiments by Kahneman and Tversky (1983) find that expressing probabilities as natural frequencies can mitigate the conjunction fallacy, while Gigerenzer and Hoffrage (1995) show that it also facilitates Bayesian reasoning. Cosmides and Tooby (1996) confirm the latter result and argue that human cognitive architecture has evolved to process natural frequencies rather than single-event probabilities in many situations. Schapira et al (2001) report that participants in their study identify frequency formats as being intuitive and easy to interpret. To illustrate the primacy of natural frequency in probability related cognition the reader may try to explain the meaning of the statement “a fair coin will come up heads with probability 0.5” to someone not fluent in mathematics *without referring to natural frequencies!*⁷

One theoretical issue that should be addressed is the assumption that the subject believes the realisations of Y to be independent, which is crucial for our results. However, statistical analysis of experimental data routinely rests upon this assumption, so we are confident that this will not be seen as a serious weakness. Furthermore, we do not require that the realisations are genuinely independent, as required for the validity of many statistical tests, but only that they are perceived to be independent by the subjects. While the assumption of independence should be borne in mind as a requirement for the validity of our results, we believe that in general it will be of little practical importance.

The assumption in the median elicitation described in Proposition 4 that every outcome is believed to have a non-zero probability of occurring could also be important. In most cases it is unlikely that subjects will see any outcome as *completely* impossible and would place some, perhaps infinitesimal, weight on it occurring. If there are outcomes that could credibly be seen as impossible however, this issue can be dealt with by simply removing these from the set of possible realizations prior to the elicitation.

One potential drawback is the possibility of weak incentives when eliciting probabilities with large n . For example, for $k = 3$, $n = 21$ and some sets of beliefs, the payment probability may be as low as 3.8%. This means P would have to be very large for the expected payoff to be of a significant size. It is an empirical question as to whether subjects will perceive the payment probability to be so low; any optimism about the accuracy of one’s own predictions would lead to an increased perception of the payment probability. Also, the all or nothing nature of the payment should focus the attention of subjects, especially in comparison to the QSR where one can often guarantee half the maximum payoff by placing equal weight on all options. In the design of an experiment

⁷ Another avenue we believe worth pursuing is the use of graphical aids. There has been a great deal of work on this in the fields of cognitive psychology and medical risk communication which could both complement and be complemented by experimental economics methodology. See, for example, references in Schapira et al (2001).

one may want to consider this trade-off between precision and strength of incentives when choosing n .

Weak incentives are not a problem in eliciting medians, however, as one can increase the probability of receiving the prize by reducing the number of realizations the payment is based on. In contrast to the case of probability elicitation, here reducing n does not result in loss of precision.

A related practical issue is that implementation of these methods could result in a large variance in payoffs between subjects, and also the total payment required for a session, which is often undesirable. With a small payment probability offset by a large prize the experimenter could be at risk of going over budget if too many subjects' predictions turn out to be accurate.

Which method to apply depends on the purpose of the elicitation, and the nature of the random variable one is considering. In any application involving expected utility theory the elicitation of probabilities and means will be necessary. However, if a measure of beliefs about a central tendency is desired, one must consider the tradeoff between the number of choices available and precision. For example, in a public goods game with an endowment of \$10 one could allow subjects to contribute only the whole endowment or nothing, or only multiples of \$5, thus reducing k to two or three.⁸ For larger choice sets or continuous variables, the methods in Section 3 allow for precise estimation of medians and other quantiles, even with small numbers of subjects. As such, belief elicitation should be considered as an integral part of designing an experiment as a whole, and not simply added as an afterthought. Mean and variance elicitation should be matched with small sets of outcomes, whereas median and quantile elicitation should be matched with outcomes that are as close to continuous as possible.

6 Conclusion

In this paper we have presented and characterized methods of belief elicitation which are extremely transparent to subjects and not dependent on restrictive assumptions about utility functions. We encourage experimentalists to use these methods in their own work, and especially to compare their empirical performance with other existing scoring rules.

References

- Bhatt M, Camerer CF (2005) Self-referential thinking and equilibrium as states of mind in games: fmri evidence. *Games and Economic Behavior* 52:424–459
- Blanco, Engelmann D, Koch AK, Norman HT (2010) Belief elicitation in experiments: is there a hedging problem? *Experimental Economics* 13(4):412–438

⁸ Internet experiments allow for very large n , and thus reasonably precise elicitation of beliefs about a larger choice set.

- Cosmides L, Tooby J (1996) Are humans good intuitive statisticians after all? rethinking some conclusions from the literature on judgement under uncertainty. *Cognition* 58:1–73
- Costa-Gomes MA, Weizsacker G (2008) Stated beliefs and play in normal-form games. *The Review of Economic Studies* 75:729–762
- Gigerenzer G, Hoffrage U (1995) How to improve bayesian reasoning without instruction: Frequency formats. *Psychological Review* 102(4):684–704
- Hurley TM, Shogren JF (2005) An experimental comparison of induced and elicited beliefs. *Journal of Risk and Uncertainty* 30(2):169–188
- Kahneman D, Tversky A (1983) Extensional versus intuitive reasoning: The conjunction fallacy in probability judgement. *Psychological Review* 90(4):293–315
- Karni E (2009) A mechanism for eliciting probabilities. *Econometrica* 77(2):603–606
- Le Coq C, Tremewan J, Wagner AK (2015) On the effects of group identity in strategic environments. *European Economic Review* 76:239–252
- Lipkus IM, Samsa G, Rimer BK (2001) General performance on a numeracy scale among highly educated samples. *Medical Decision Making* 21:37–44
- Manski CF (2004) Measuring expectations. *Econometrica* 72(5):1329–1376
- Offerman T, Sonnemans J, Van de Kuilen G, Wakker PP (2009) A truth serum for non-bayesians: Correcting proper scoring rules for risk attitudes. *The Review of Economic Studies* 76(4):1461–1489
- Schapira MM, Nattinger AB, McHorney CA (2001) Frequency or probability? a qualitative study of risk communication formats used in health care. *Medical Decision Making* 21:459–467
- Schlag KH, Tremewan J, Van der Weele JJ (2015) A penny for your thoughts: A survey of methods for eliciting beliefs. *Experimental Economics* 18(3):457–490
- Schotter A, Trevino I (2014) Belief elicitation in the laboratory. *Annu Rev Econ* 6(1):103–128
- Schwarz LM, Woloshin S, Black WC, Welch HG (1997) The role of numeracy in understanding the benefit of screening mammography. *Annals of Internal Medicine* 127:966–971
- Wilcox NT, Feltovich N (2000) Thinking like a game theorist: Comment, mimeo, University of Houston